Chemoenzymatically prepared heparan sulfate containing rare 2-O-sulfonated glucuronic acid residues

Search within citing articles

Glycosaminoglycan microarrays for studying glycosaminoglycan–protein systems

JE Chittum, A Thompson, UR Desai - Carbohydrate Polymers, 2024 - Elsevier

More than 3000 proteins are now known to bind to glycosaminoglycans (GAGs). Yet, GAG–protein systems are rather poorly understood in terms of selectivity of recognition, molecular …

☆ Save 99 Cite Cited by 12 Related articles All 5 versions

Heparan sulfate microarray reveals that heparan sulfate–protein binding exhibits different ligand requirements

C Zong, A Venot, X Li, W Lu, W Xiao… - Journal of the …, 2017 - pmc.ncbi.nlm.nih.gov

Heparan sulfates (HS) are linear sulfated polysaccharides that modulate a wide range of physiological and disease-processes. Variations in HS epimerization and sulfation provide …

☆ Save 99 Cite Cited by 142 Related articles All 9 versions

Synthetic heparin and heparan sulfate: probes in defining biological functions

CT Tsai, MML Zulueta, SC Hung - Current opinion in chemical biology, 2017 - Elsevier

Highlights•Heparin and heparan sulfate binds various proteins and modulate their activity.•Advances allowed the chemical preparation of long sugars and sugar libraries.• …

☆ Save 99 Cite Cited by 42 Related articles All 6 versions

A Hexasaccharide Containing Rare 2-*O*-Sulfate-Glucuronic Acid Residues Selectively Activates Heparin Cofactor II

NV Sankaranarayanan, TR Strebel… - Angewandte Chemie …, 2017 - Wiley Online Library

Glycosaminoglycan (GAG) sequences that selectively target heparin cofactor II (HCII), a key serpin present in human plasma, remain unknown. Using a computational strategy on a …

☆ Save 99 Cite Cited by 70 Related articles All 19 versions

Discovery of rare sulfated *N*-unsubstituted glucosamine based heparan sulfate analogs selectively activating chemokines

P Jain, CD Shanthamurthy, S Leviatan Ben-Arye… - Chemical …, 2021 - pubs.rsc.org

Achieving selective inhibition of chemokines with structurally well-defined heparan sulfate (HS) oligosaccharides can provide important insights into cancer cell migration and …

☆ Save 99 Cite Cited by 26 Related articles All 8 versions

Allosteric partial inhibition of monomeric proteases. Sulfated coumarins induce regulation, not just inhibition, of thrombin

S Verespy III, AY Mehta, D Afosah, RA Al-Horani… - Scientific Reports, 2016 - nature.com

Allosteric partial inhibition of soluble, monomeric proteases can offer major regulatory advantages, but remains a concept on paper to date; although it has been routinely …

☆ Save 99 Cite Cited by 47 Related articles All 8 versions

A simple method for discovering druggable, specific glycosaminoglycan-protein systems. Elucidation of key principles from heparin/heparan sulfate-binding proteins

A Sarkar, UR Desai - PloS one, 2015 - journals.plos.org

Glycosaminoglycans (GAGs) affect human physiology and pathology by modulating more than 500 proteins. GAG-protein interactions are generally assumed to be ionic and …

☆ Save 99 Cite Cited by 56 Related articles All 20 versions »

Rational designing of glyco-nanovehicles to target cellular heterogeneity

P Jain, CD Shanthamurthy, PM Chaudhary… - Chemical Science, 2021 - pubs.rsc.org

The aberrant expression of endocytic epidermal growth factor receptors (EGFRs) in cancer cells has emerged as a key target for therapeutic intervention. Here, we describe for the first …

☆ Save 99 Cite Cited by 23 Related articles All 9 versions

Sulfation of Heparan and Chondroitin Sulfate Ligands Enables Cell-Specific Homing of Nanoprobes

S Mardhekar, B Subramani, P Samudra… - … A European Journal, 2023 - Wiley Online Library

Demystifying the sulfation code of glycosaminoglycans (GAGs) to induce precise homing of nanoparticles in tumor cells or neurons influences the development of a potential drug-or …

☆ Save 99 Cite Cited by 14 Related articles All 9 versions

2-O, 3-O desulfated heparin blocks high mobility group box 1 release by inhibition of p300 acetyltransferase activity

S Zheng, AB Kummarapurugu… - American journal of …, 2017 - academic.oup.com

High mobility group box 1 (HMGB1) is an alarmin released from macrophages after infection or inflammation and is a biomarker of lung disease progression in patients with cystic …

☆ Save 99 Cite Cited by 36 Related articles All 8 versions